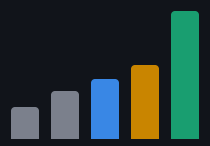
HIT SONG PREDICTION · TECHNION · PROJECT IN AI

What really predicts a hit?

The story of our project: what we tried, what worked, and what turned out to be a trap.

Lior Shalon & Nathanael Uzzan

Technion – Israel Institute of Technology · Project in Artificial Intelligence



THE QUESTION

Can a computer guess, before a song comes out, whether it will be a hit?

What is a hit?

A song that entered the **Billboard Hot 100**, the weekly chart of the most popular songs in the USA.

How the computer learns

We show it thousands of songs whose answer we already know. It finds patterns. Then we test it on songs it has **never seen**.

What is a good score?

Guessing at random gives **50 %** right, like tossing a coin. Everything we do is compared to that.

DATA

Four datasets — the definition of "hit" changes every time

DATASET	SONGS	HITS	HIT LABEL	HOW NON-HITS WERE CHOSEN
Spotify Hit Predictor (Kaggle, 1960–2019)	40,560	49 %	Weekly Hot 100 in that decade	Rules: artist *and* genre absent from the hit list
audio subset (30 s Deezer clips)	3,562	52 %	same	same
ISMIR 2019 (Zangerle et al., peer-reviewed)	11,662	50 %	Hot 100 match	Random Million Song Dataset tracks
SpotGenTrack	101,909	19 %	Spotify popularity ≥ 55	Everything else
~~Billboard year-end 2000–2023~~	3,397		DROPPED 2,911 rows have no audio features	

Paper · Table I

What goes in, what comes out

Song audio
Spotify Web API

Release date
which decade

Artist name
+ Billboard Hot 100 since 1958

→

13 sound numbers
danceability, energy, loudness, acousticness, …

6 decade boxes
60s 70s 80s 90s 00s 10s — one is 1

1 artist-history number
log(1 + Hot 100 songs in **earlier** decades)

→

Random Forest
80 decision trees. Each asks yes/no questions about the 20 numbers and votes. 8 inputs tried per question. **Never tuned.**

→

Output
P(hit) between 0 and 1

≥ 0.5 → **hit**
< 0.5 → **not a hit**

Input: one song = 20 numbers, all known on the day it comes out. **Output:** will it enter the Billboard Hot 100?
Learned from 32,000 songs whose answer we know; **tested on** 8,000 songs it never saw.

Paper · Fig. 1

STEP 1

We started with the sound

Spotify describes every song with **13 numbers**: how danceable, how energetic, how fast, how loud, how acoustic, how happy it sounds…

`danceability` `energy` `tempo` `loudness` `acousticness` `valence (happy)` `+ 7 more`

78%

right, on 40,560 songs (half hits, half not). Much better than a coin, but **1 song in 5 is still wrong**.

Vogue (Madonna, 1990), a song the model never saw

1 · Raw data

danceability **0.737**
energy **0.880**
loudness **−6.4 dB**
acousticness **0.004**
duration **5.3 min** … (13 values)

file: **1990s**
artist: **Madonna**

→

2 · 20 inputs

13 sound values, unchanged

60s	70s	80s	90s	00s	10s
0	0	0	1	0	0

Madonna: **19** Hot 100 songs before 1990 → h = log(1+19) = **3.0**

→

3 · Each tree asks

`danceability ≤ 0.49?` **no**
`acousticness ≤ 0.76?` **yes**
`duration ≤ 5.7 min?` **yes**
`artist history ≤ 0.35?` **no**
`loudness ≤ −6.9 dB?` **no**
`… 4 more …`
`→ leaf: 646 songs,` **100 % hits**

→

4 · Vote

80 / 80 trees say hit

P(hit) = **1.00**

→ **HIT**
(true: no. 1)

Every song goes through the same 4 steps. On the 8,113 test songs of this split, the answer is right **83.3 %** of the time. Training: each tree learns on its own random sample of the 32,447 training songs and, at each question, picks the input and threshold that best separate hits from non-hits.

Paper · Fig. 2 · tree 1 of the trained forest

METHOD · HOW EVERY GAIN IS TESTED

Two estimates for every comparison — and a test that is not fooled by overlap

1 · Canonical split

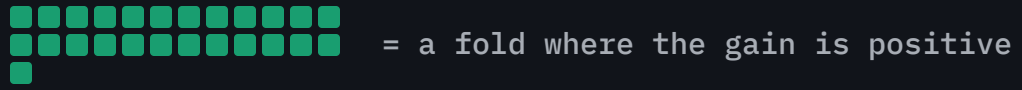
The split of our earlier experiments (seed 42; 20,000 train / 2,000 test). Two models compared song by song with **McNemar's test**.



Reproduces the old numbers exactly: .773 → .822.

2 · Repeated cross-validation

5 folds × 5 repeats = **25 test sets**. Mean, 95 % confidence interval, and the **Nadeau–Bengio corrected t-test** — the naive t-test is over-confident because the folds overlap.



Scaling, PCA and title models are fitted inside each training fold only.

What is the decade input, and how do we build it?

What it is

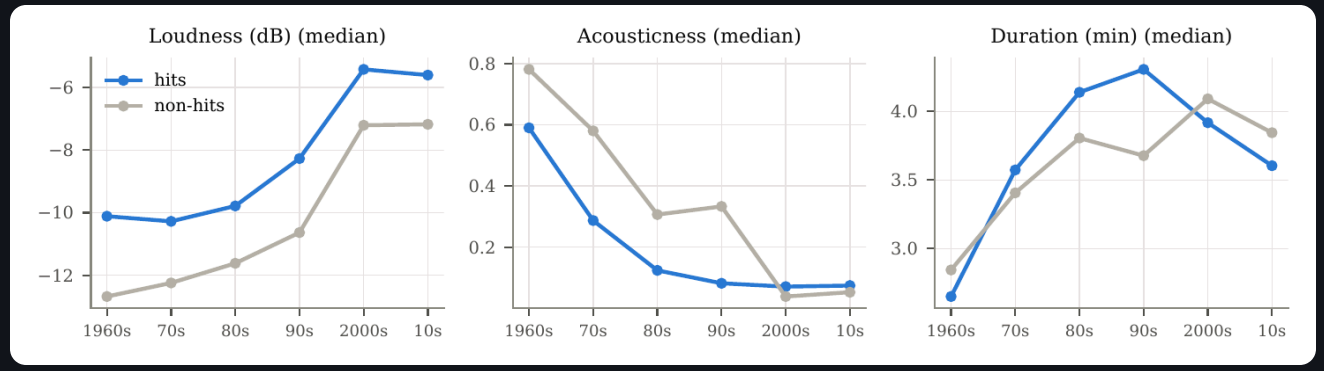
The decade in which the song came out: 1960s, 70s, 80s, 90s, 2000s or 2010s.

How we build it

The dataset comes as **6 files**, one per decade (`dataset-of-60s.csv` … `dataset-of-10s.csv`). The decade = the file. We turn it into **6 yes/no boxes**, exactly one is 1 (one-hot), because a decade is a category, not a number.

Why it helps

Each decade is half hits, so alone it is a coin toss (50.3 %). But **what a hit sounds like changes with the era** (right): with the decade, the trees learn one rule per era. **+2.0 points**.



Median of three sound numbers for hits (blue) and non-hits (gray). A 2000s non-hit is louder than any 1960s–80s hit; hits were less acoustic than non-hits until the 1990s, slightly more after; hits were longer in the 1970s–90s, shorter after 2000.

Paper · Sec. IV-B, Fig. 3

What is artist history, and how do we compute it?

Rule

For a song, take its artist and count their **different songs on the Billboard Hot 100 in earlier decades**. Never the same decade, never the future, so it is known on release day.

Formula

h = log(1 + count). The log squeezes big counts: 40 earlier hits is not twice 20.

For every artist

Names are cleaned (lower case, accents, “feat.” removed) and matched to the full Hot 100 history (1958–2026). Computed separately for every artist and every decade.

EXAMPLE · MADONNA

1980s: 19 Hot 100 songs

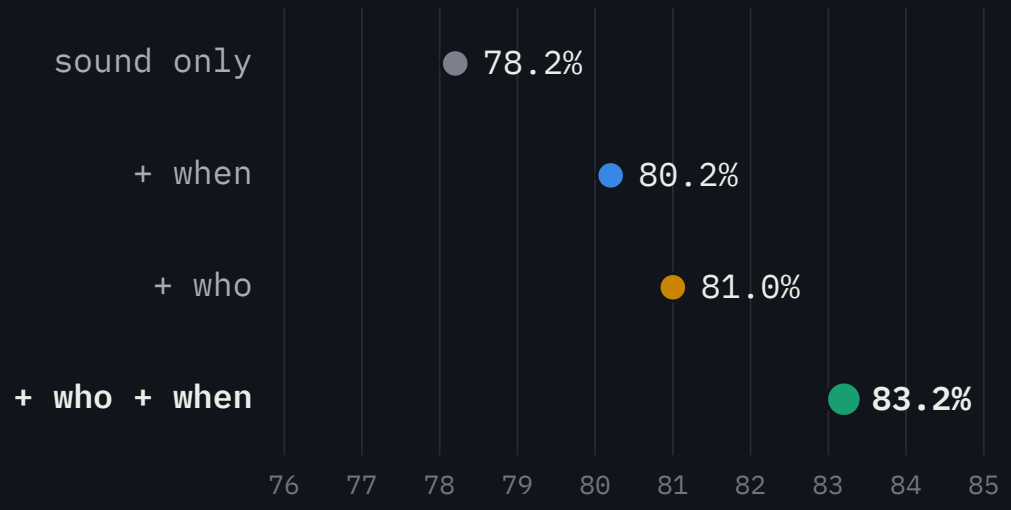


SONG	DECADE	EARLIER SONGS	H
Holiday	1980s	0	0.0
Vogue	1990s	19	**3.0**
Hung Up	2000s	44	3.8

A brand-new artist always gets **h = 0**.

Paper · Eq. (1), Fig. 4

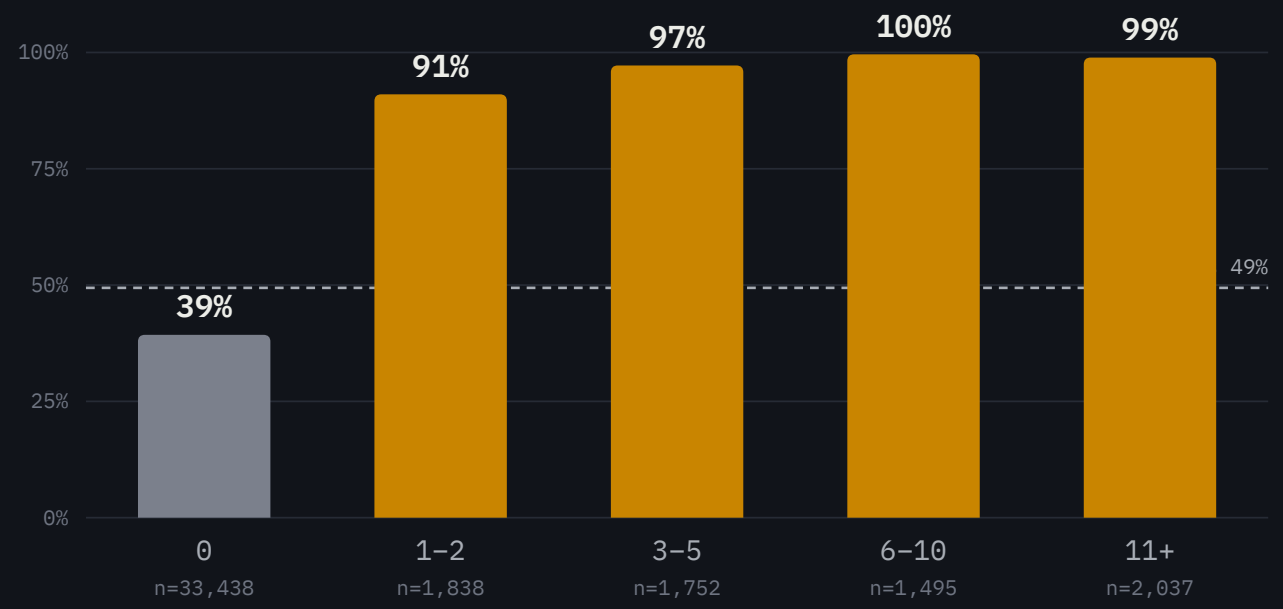
From 78 % to 83 % right



+5 points

Better in **25 out of 25** tests. Each clue helps on its own; together they help the most.

One earlier hit changes everything



Share of songs that are hits, by the artist's number of earlier hits

Artist with **no** earlier hit: **39 %** of their songs are hits.

With **1 or 2** earlier hits: **91 %**.

STEP 3

Is it luck? We tried to break it — it held

5 different algorithms

Random Forest, XGBoost, SVM, k-NN, logistic regression: all gain **+4.6 to +5.7**. It is the clues, not the algorithm.

Every decade

From the 1960s to the 2010s: **+3.7 to +6.6** every time.

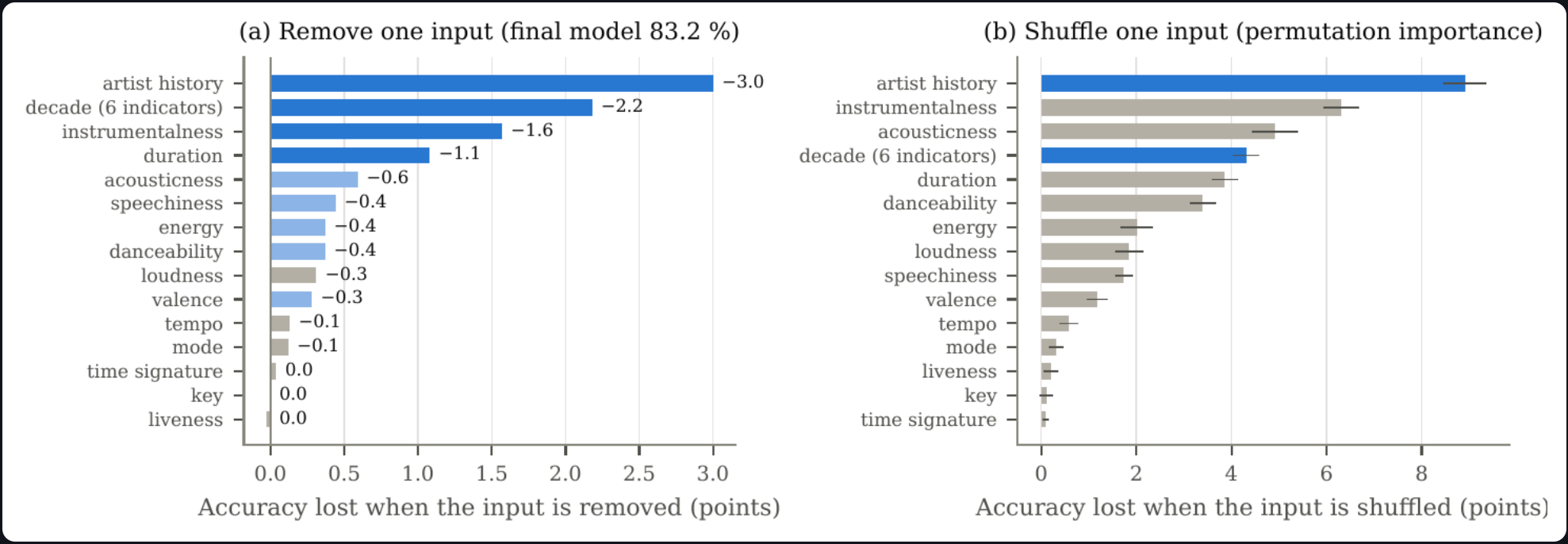
The future

Trained on old decades, tested on a decade it never saw: still **+1.9 to +3.4**.

The only other study

We redid its exact method (Du, 2022) and added our clue: **better in 50 out of 50** tries.

Where does the accuracy come from?



Remove and retrain: artist history **−3.0**, decade **−2.2**, instrumentalness −1.6, duration −1.1 (all significant). Key, liveness, mode, tempo: 0.

Shuffle in the trained model: same top. Duration matters only through the era.

Alone: sound 78.2 %, artist 67.0 %, decade 50.3 %. **Together 83.2 %**: they complete each other.

Paper · Sec. V-B, Fig. 7

Putting every input in one model: higher score, wrong reasons

MODEL (35,370 SONGS)	ACCURACY
Final model (20 inputs)	82.5 %
Everything (8 blocks)	**88.3 %**
– song title	84.3 % (−4.0)
– artist history	86.4 % (−1.9)
– 13 sound numbers	87.1 % (−1.2)
– FFT / – market trend	87.9 % (−0.4 / −0.3)
– decade	88.1 % (−0.2, n.s.)
– title length / – chorus_hit	88.3 % (0)

Most of the extra comes from the **song title**, which only reveals how the non-hits were collected.

The rest comes from FFT and market trend, which fail the genre and era tests (chapter 5). With them in, the decade is redundant: all three describe the era.

PCA of the 13 sound numbers: replacing them by components only loses accuracy (−1.7), so we keep the raw numbers (appendix).

→ We keep the 20 inputs.

Paper · Table VI, Fig. 11

A trap: how the non-hits were chosen

The dataset's own rules: a non-hit must come from an artist who **had no hit** that decade, in a **niche genre** (jazz, Latin, classical…).

So “the artist had hits before” is partly the dataset's rule, not the music.

97%

of songs by an artist with an earlier hit are hits **in this dataset**. Our +5 is too optimistic here.

STEP 4 · THE FAIR TEST

Same year for hits and non-hits: the date is useless, the artist still helps

The year

useless

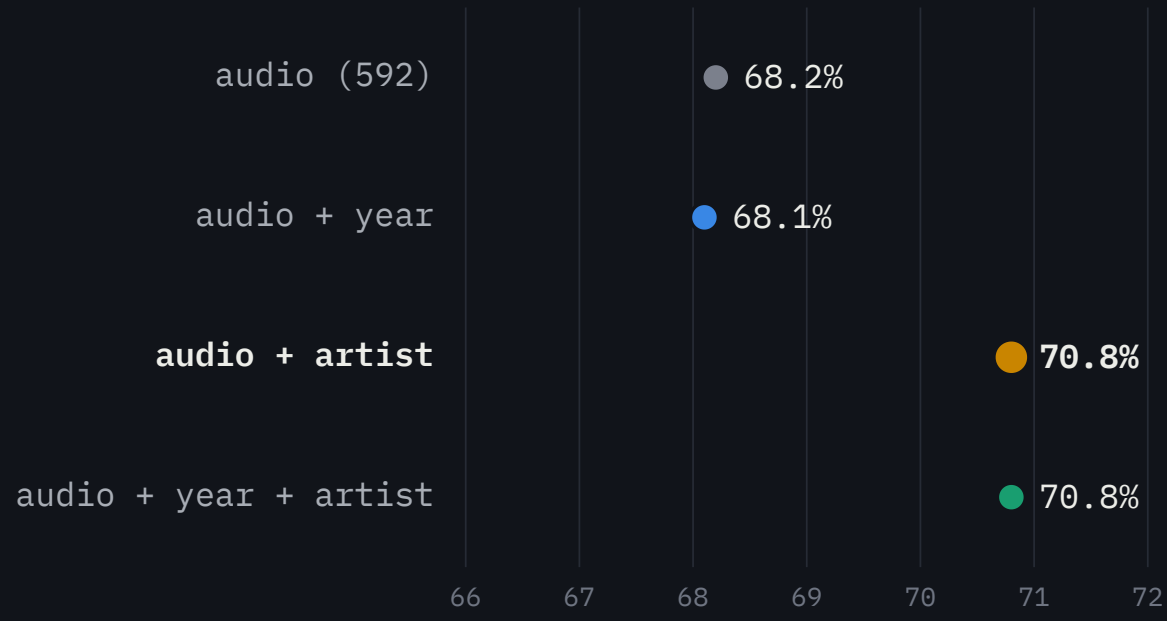
It was only a trick of the dataset.

The artist's history

+2.6 points

Better in 10 out of 10 tests. **This is our honest number**: smaller than +5, but real.

Match the years: release year becomes useless, artist history still adds +2.6



Same data, one non-hit from the same year for every hit · 5×2 CV · paper Table VI

+2.6

points, $p = 3{\cdot}10^{-5}$, 10/10 folds. Negatives chosen **neither by artist nor by date**.

This is the number we defend as the transferable value of artist history. Year: p = .66; year alone falls to 44.8 %, below chance.

TWO DIFFERENT QUESTIONS

“Unseen” artist ≠ “new” artist

Unseen artist — not in training

The model never trained on this artist's songs, but the artist may have **earlier hits** (h > 0). We test it by putting each artist only in training *or* only in test.

Result: the gain **stays** — +5.0 on the big dataset, +2.9 on the fair ISMIR test. So the model does not just recognise artists.

New artist — no earlier hit

h = 0 by definition, so artist history **cannot help** (+0.4, not significant). Only the **decade** helps them (+2.2).

On the big dataset, the whole gain of artist history comes from the 18 % of songs by known artists — 97 % of them are hits.

Paper · Sec. IV-D and V-E

HOW WE DECIDED WHAT TO KEEP

Every input went through the same four questions

1 · Known on release day?
No future, no own period.
✗ genre tags (assigned later)

→

2 · Gain under 25 tests?
corrected p < 0.05
✗ title length, chorus_hit, librosa, trend

→

3 · Survives the controls?
same years · same genres · unseen artists
✗ titles, FFT, release year

→

4 · Survives Bonferroni?
p < 0.0008 over ~60 tests
~ small lyrics test (SpotGenTrack)

→

Kept
decade
artist history
lyrics (where they exist)

Genre and FFT were the strongest-looking inputs after the titles. We kept testing them until we understood why: they recognise the niche genres from which the dataset took its non-hits.

Paper · Fig. 5

We tried more clues — and checked every gain

Song titles: +6.7

Too good to be true. The titles reveal **how the dataset was collected**, not what makes a hit.

✗ REMOVED

Detailed sound (185 measures)

+3.8 on one test… but only **+0.9 over 25 tests**, which is nothing. The first result was luck. ✗ REMOVED

Small details

Length of the title, where the chorus starts, number of sections: **nothing**. ✗ REMOVED

SURVIVAL RANKING · EVERY P-VALUE

What survived, what was a shortcut, what did not survive

SURVIVED · LOW p + CONTROL PASSED

Decade + artist, big dataset: +5.0 · $p = 7{\cdot}10^{-17}$

Every classifier: +4.6 to +5.7 · $p \le 4{\cdot}10^{-5}$

Du's models + artist: 50/50 splits · $p \le 6{\cdot}10^{-6}$

Future decades: +1.9 to +3.4 · $p \le 10^{-7}$

Artist, fair ISMIR test: +2.6 · $p = 3{\cdot}10^{-5}$

Lyrics, ISMIR, all controls: +4.4 · $p = 3{\cdot}10^{-5}$

First 3 weeks + context: AUC +.047 · $p < 10^{-6}$

SHORTCUT · LOW p, CONTROL FAILED

Song title: +6.7 · $p = 4{\cdot}10^{-20}$ → artefact

Artist genre: +10.1 · $p = 8{\cdot}10^{-23}$ → assigned later

FFT: +1.4 · $p = 4{\cdot}10^{-9}$ → +0.1 when genres match

Release year, ISMIR: +3.1 · $p = 2{\cdot}10^{-4}$ → −0.1 when years match

DID NOT SURVIVE · HIGH p

Lyrics, SpotGenTrack: p = .002 (indicative)

Market trend, per feature: p = .006

FFT on top of artist, subset: p = .008

Decade, small subset: p = .03

Rich librosa: p = .15 (one split said .002)

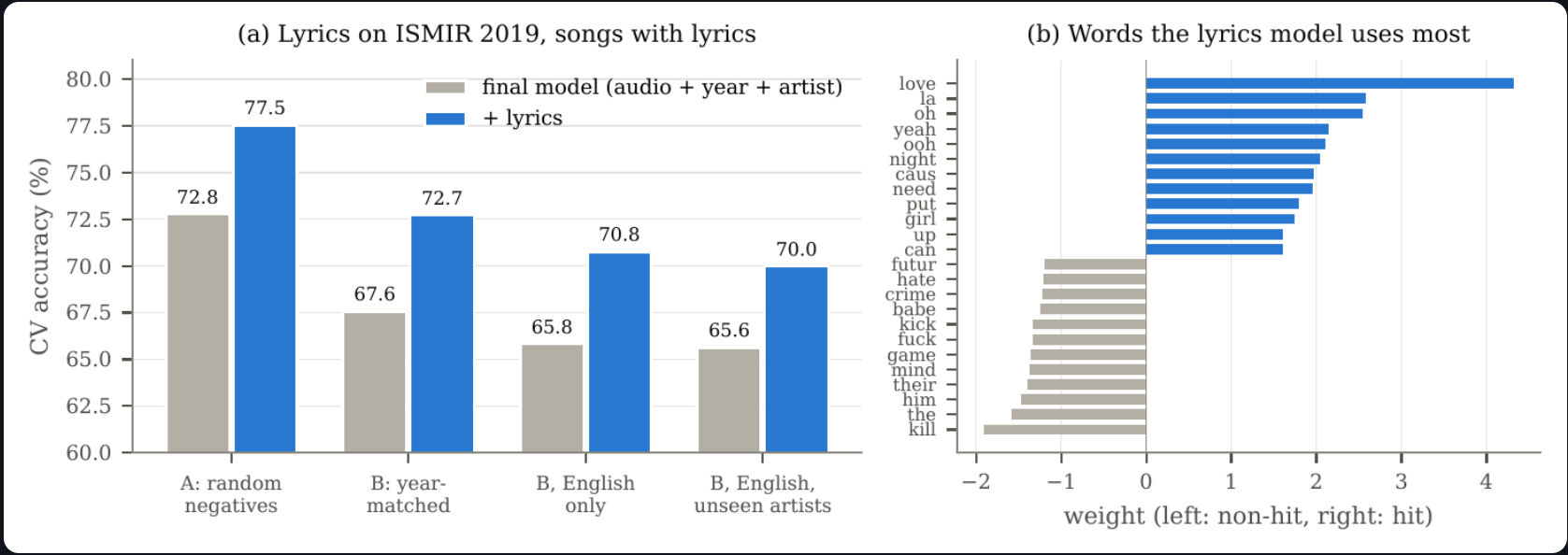
Artist history for new artists: p = .17

Title length .28 · trend .66 · chorus_hit .82

Survives = significant under 25 tests, still significant after Bonferroni ($p < 8{\cdot}10^{-4}$ over ~60 tests), and passes the control that removes its dataset's known shortcut.

Paper · Table VII

The lyrics: the only content input that survives every test



Inputs: 8 word statistics (length, richness, “I”, “you”, “love”…) + a bag-of-words model trained inside each fold.

Traps removed: only songs with lyrics on both sides; English-only; same years; no artist in both training and test.

+4.4 points after all controls (10/10, $p = 3{\cdot}10^{-5}$). Hits sing *love, oh, yeah*; non-hits *kill, hate, crime*: partly mainstream themes.

Paper · Sec. V-G, Fig. 9

Can time series predict a hit? It depends on *when* you ask

BEFORE the song comes out (our question)

Does the song sound like the recent hits? **+0.1, nothing**: the trend only knows the decade, which we already had. ✗ REMOVED

No

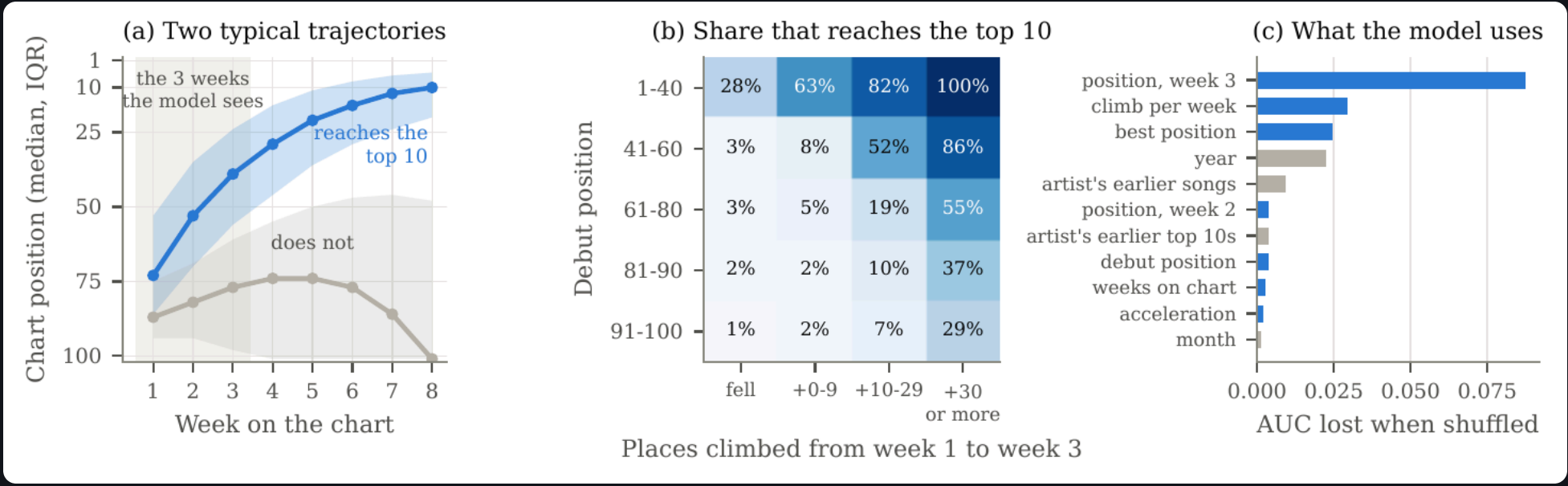
AFTER it comes out (another question)

The song just entered the chart. Its **first 3 weeks** tell if it will reach the **top 10**: right 89 times out of 100 when comparing two songs (AUC .89). ✓ KEPT

Yes

Both are true. Time series do not help guess a hit *before* it is out — but once it is out, they are the best clue we found.

What happens in the first 3 weeks decides a top-10 hit



First k weeks = the song's chart position in each of its first k weeks (101 = dropped off). *Vogue*: 39 → 28 → 12, then no. 1.

Paper · Sec. V-H, Fig. 10

Future top-10s climb fast: median 73 → 53 → 39. The others stay near 80. Debut 61–80 and falling: **3 %**; climbing 30+ places: **55 %**.

The model uses the **week-3 position** and the **climb**, not the debut. AUC **.89** with 3 weeks, .59 without.

CONTRIBUTIONS

What we did, how, and what changed compared with before

CONTRIBUTION	HOW WE DO IT	BEFORE	OURS
Artist history + era	earlier-decade Hot 100 count; decade one-hot; 25 tests	superstar flag, no test (Interiano 2018); sound 78.2 %	**83.2 %** (+5.0, 25/25)
Head-to-head	Du's exact method + our input, 50 splits	Du 2022: 84 % / 82 %	**86.7 % / 86.1 %**, 50/50
External check	ISMIR 2019, then same-year non-hits	best published 75.04 %	**76.3 %**; fair test +2.6
Unseen / new artists	artist-grouped tests; split by earlier hit	not tested	gain kept (+5.0, +2.9); new artists: era only
Ablation	remove / shuffle one input; everything at once	not reported	artist −3.0, decade −2.2; “everything” = title artefact
Audit of shortcuts	25 tests, genre- and year-matched controls	one split, no control	titles, FFT, year, librosa rejected
Lyrics	musiXmatch words, all controls	lyrics in models on other data	**+4.4** ($p = 3{\cdot}10^{-5}$)
First weeks	first 3 chart positions + climb	early views predict online content	AUC **.89** (.59 without)

Paper · Table I

Our final model: one Random Forest, 20 numbers per song

13 Spotify sound descriptors + **6** decade boxes (when) + **1** artist history (who)

Artist history = log(1 + the artist's Hot 100 songs in **earlier decades only**).

→ **Random Forest**: 80 trees, 8 features per question, **never tuned** → hit / not hit

Same setting as the reference paper, so results compare. XGBoost is 1 point better on average, not significantly (p = .87).

OTHER DATASETS, SAME RECIPE

Their own sound descriptors (592 Essentia on ISMIR) + year or decade + artist history; Random Forest, 200 trees.

NOT IN THE MODEL

Titles, detailed sound, FFT, genre, market trend, chorus_hit / sections: traps or noise. **Lyrics**: added where they exist (ISMIR: 77.5 %).

A SEPARATE MODEL, AFTER RELEASE

Songs already on the chart: first 3 weeks (positions, best rank, slope, acceleration) + year, month, artist history → Random Forest → top 10 or not. AUC .89.

SCOREBOARD

How much our model improved each dataset

DATASET	MEASURE	WITHOUT OUR CLUES	WITH OUR MODEL	GAIN	COMPARED WITH PUBLISHED WORK
Spotify Hit Predictor · 40,560 songs	accuracy	78.2 %	**83.2 %**	+5.0	F1 of hits .787 → .831
Same data, Du (2022) method · 2000s	accuracy	84.6 %	**86.7 %**	+2.1	Du published 84 % · we win 50/50 splits
Same data, Du (2022) method · 2010s	accuracy	82.9 %	**86.1 %**	+3.2	Du published 82 % · we win 50/50 splits
ISMIR 2019 · as in the original study	accuracy	73.9 %	**76.3 %**	+2.4	best published **75.04 %** → beaten
ISMIR 2019 · fair test (same years)	accuracy	68.2 %	**70.8 %**	+2.6	our honest number
ISMIR 2019 · + lyrics, fair test, unseen artists	accuracy	65.6 %	**70.0 %**	+4.4	songs with lyrics, English only
SpotGenTrack · 101,909 tracks (19 % hits)	AUC	.690	**.777**	+.087	hits found 8 % → 25 %
Hot 100, after release · 31,925 songs	AUC (top 10)	.59	**.89**	+.30	with the first 3 weeks on the chart

“Without our clues”: the same model on the dataset's usual inputs (13 Spotify descriptors; Du's own inputs; 592 Essentia descriptors + year on ISMIR; year, month and artist history for the Hot 100). Cross-validated means.

SUMMARY

What we proved

TRUE — IT SURVIVED EVERY TEST

1 · **Who sings it matters**: +5 points on the big dataset, **+2.6 on a fair test**.

2 · We **beat the published results**: 76.3 % vs 75.0 %, and the other study 50 times out of 50.

3 · It works with **any algorithm** and for **artists never seen**.

4 · For a **brand-new artist**, only the era helps a little.

5 · **Lyrics** add **+4.4** on a fair test, with unseen artists.

6 · **After** a song comes out, its first 3 weeks predict a top-10 hit (AUC .89). **Before**, time series add nothing.

FALSE — IT WAS A TRAP

Song titles: reveal how the data was collected.

Detailed sound: the gain was luck.

FFT: it was recognising the genre.

The year: hits were just older songs.

Music trends: they only know the decade.

In the sound itself, nothing survived every test — the lyrics did.

A hit is mostly who sings it, when and what it is about — and, once it is out, how its first weeks go.

Everything is public

CODE · EVERY NUMBER, ONE SCRIPT EACH

github.com/uzznat/hit-song-prediction

`repro/` scripts → `paper_results/` tables → `paper/main.pdf`. Seeds fixed. `docs/DATA.md` says where each dataset goes.

DATA

Spotify Hit Predictor · github.com/fortyTwo102/The-Spotify-Hit-Predictor-Dataset (Kaggle: theoverman)

Billboard Hot 100 · github.com/utdata/rwd-billboard-data

ISMIR 2019 · zenodo.org/records/3258042

SpotGenTrack · data.mendeley.com/datasets/4m2x4zngny

musiXmatch lyrics · millionsongdataset.com/musixmatch

Audio previews · Deezer public API

Thank you — questions?

Lior Shalon & Nathanael Uzzan · every number comes from the scripts in `repro/` and the paper.

APPENDIX

The details behind every step

Algorithms and tests used, every check of the main result, the trap in the data, the external datasets, the audit, the time-series study, the statistics and the corrections.

How the computer learns, and how we test it

A Random Forest = 80 small decision trees

Each tree asks simple questions — *is it danceable? is it long? is it acoustic?* — and votes “hit” or “not hit”. The majority wins. We never tune it, so any improvement comes from the clues we give it, not from tweaking.

We test it 25 times

We hide 1/5 of the songs, train on the rest, and guess the hidden ones. Then again with other hidden songs: **25 tests** in total. An improvement counts only if it appears again and again, and passes a statistical test.

One main model, four others to check that the result is not tied to it

ALGORITHM	WHY WE USE IT	AUDIO	+ DECADE + ARTIST	VERDICT
Random Forest 80 trees, 8 features per split	Configuration of the reference paper (Middlebrook & Sheik), so results compare; learns non-linear rules (decade × audio). **Never tuned**: a gain must come from the features.	78.0	83.2	KEEP main model for every experiment
XGBoost	The strongest learner for tabular data.	78.8	84.5	Best average, but ties the Random Forest on the canonical split (p = .87). Reported, not needed.
SVM (RBF kernel)	A kernel method, very different from trees.	77.3	82.8	Same gain; slower, no probabilities.
k-NN (k = 25)	Distance-based: “a song is a hit if similar songs were”.	75.0	79.6	Same gain; weaker overall.
Logistic regression	Linear and interpretable; also one of Du’s models.	72.7	78.2	Weakest: the relations are not linear. Kept for the head-to-head.

All five gain **+4.6 to +5.7 points**: the improvement comes from the features, not the model. External datasets use a 200-tree Random Forest. 5-fold CV · Table IV

Feature extraction, explanation and statistics

TURNING AUDIO INTO FEATURES

librosa: 185 descriptors from 30-second clips (timbre, rhythm, harmony).

FFT: 31 spectral descriptors (energy per frequency band).

PCA: compresses the 185 librosa descriptors to 40 components.

Essentia: the 592 descriptors of the ISMIR dataset.

TEXT AND EXPLANATION

TF-IDF + logistic regression: turns the song title into one probability, trained out-of-fold so the test songs are never seen.

TreeSHAP: shows which features the Random Forest relies on: artist history #1, decade #2, instrumentalness #3.

TESTING EVERY GAIN

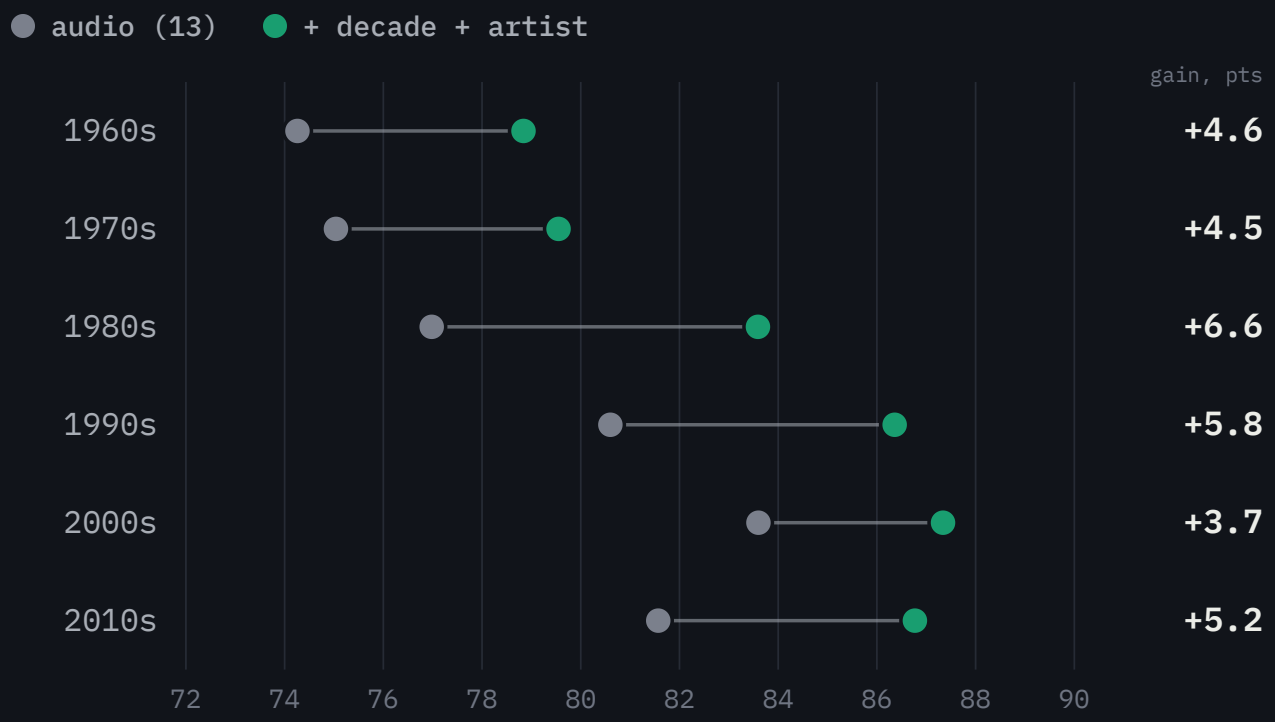
McNemar: compares two models song by song on one split.

Corrected t-test (Nadeau–Bengio): compares two models over 25 folds without over-confidence.

Bonferroni: ~60 tests, so a result needs $p < 8{\cdot}10^{-4}$ to count as solid.

ROBUSTNESS · EVERY DECADE

The gain holds in all six decades, from +3.7 to +6.6 points



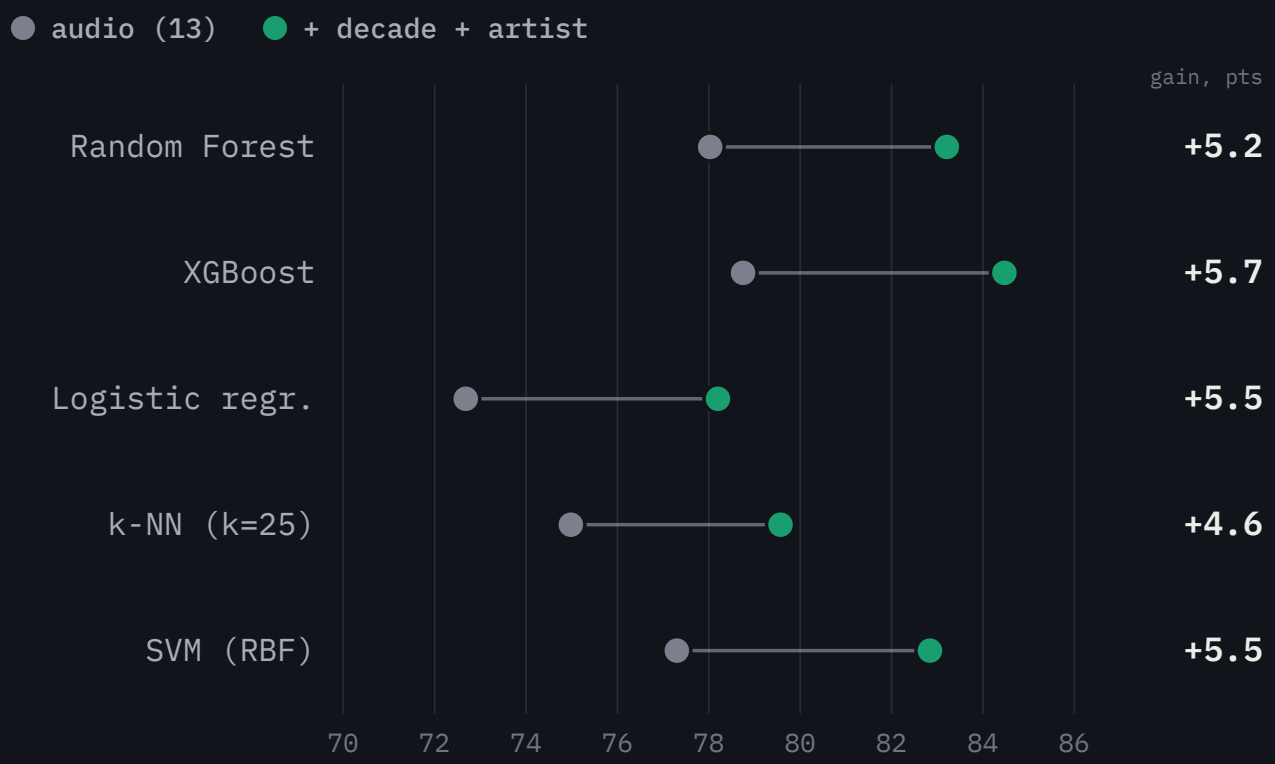
5×5 CV accuracy per decade · paper Table III

Each decade file is ~50 % hits, so the decade alone says nothing.

Its gain can only come from **interactions with the audio** — the model learns what a hit sounded like in that era.

ROBUSTNESS · EVERY CLASSIFIER

Five very different models, the same +5 points: it is the features, not the model



5-fold CV, corrected t-test, all 5 folds improve · paper Table IV

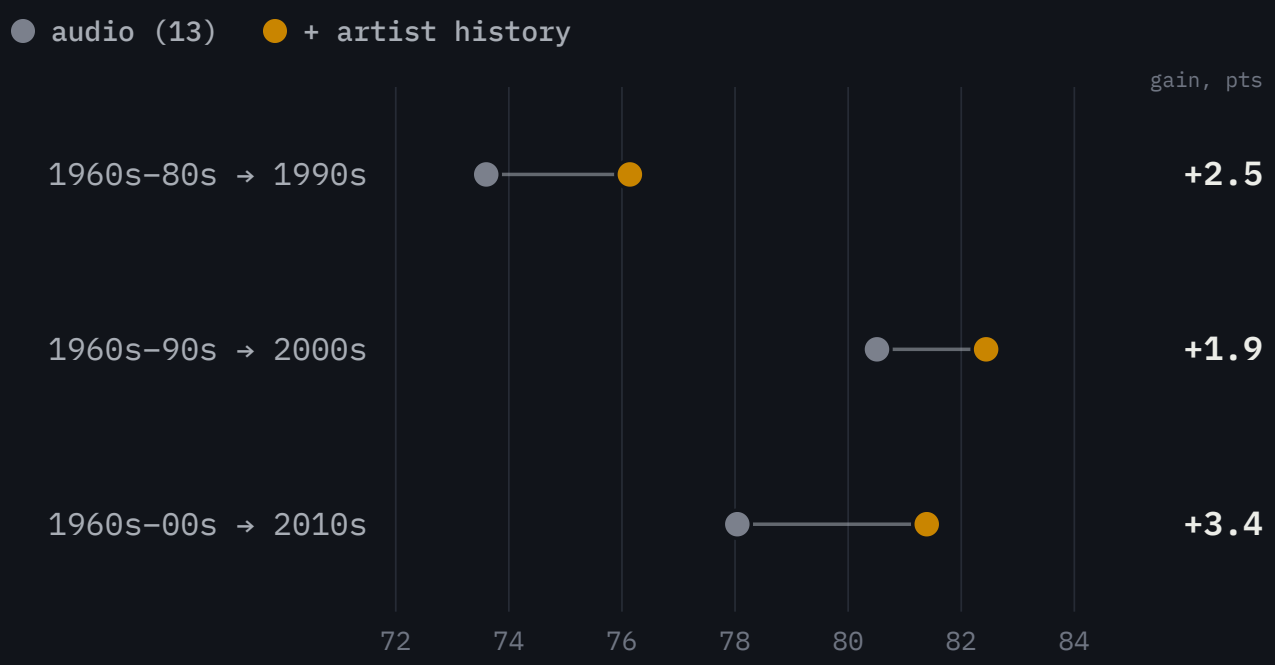
84.5%

XGBoost is the strongest on average — but on the canonical split it ties with the Random Forest (.820 vs .822, p = .87).

We keep the **Random Forest** as the main model: comparable with prior work, untuned, and within noise of XGBoost.

TreeSHAP ranks **artist history #1**, the six decade indicators #2 and instrumentalness #3.

Train on the past, predict a decade the model has never seen

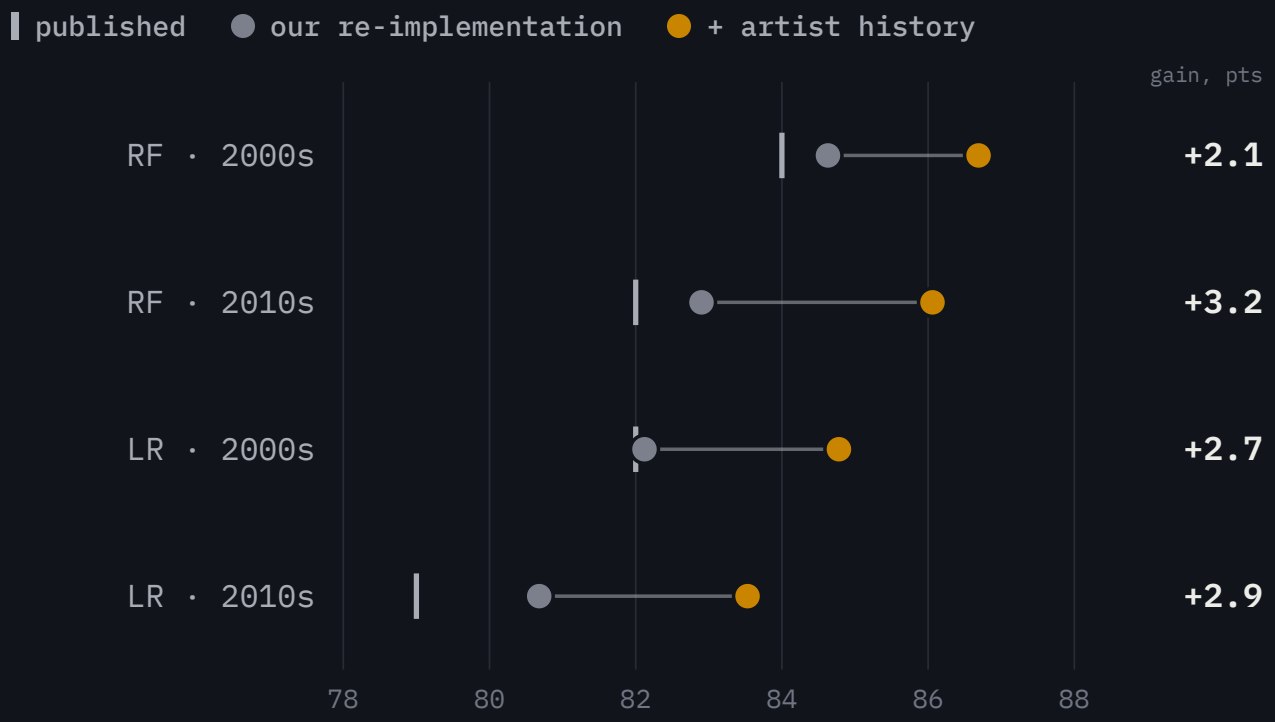


Random Forest, trained on all earlier decades · McNemar p between 10^{-7} and 10^{-21}

Artist history still adds **+1.9 to +3.4 points** on unseen decades, with similar gains for XGBoost.

The decade cannot be used here: the test decade never appears in training.

Their exact models and inputs, plus our feature: better on 50 out of 50 splits

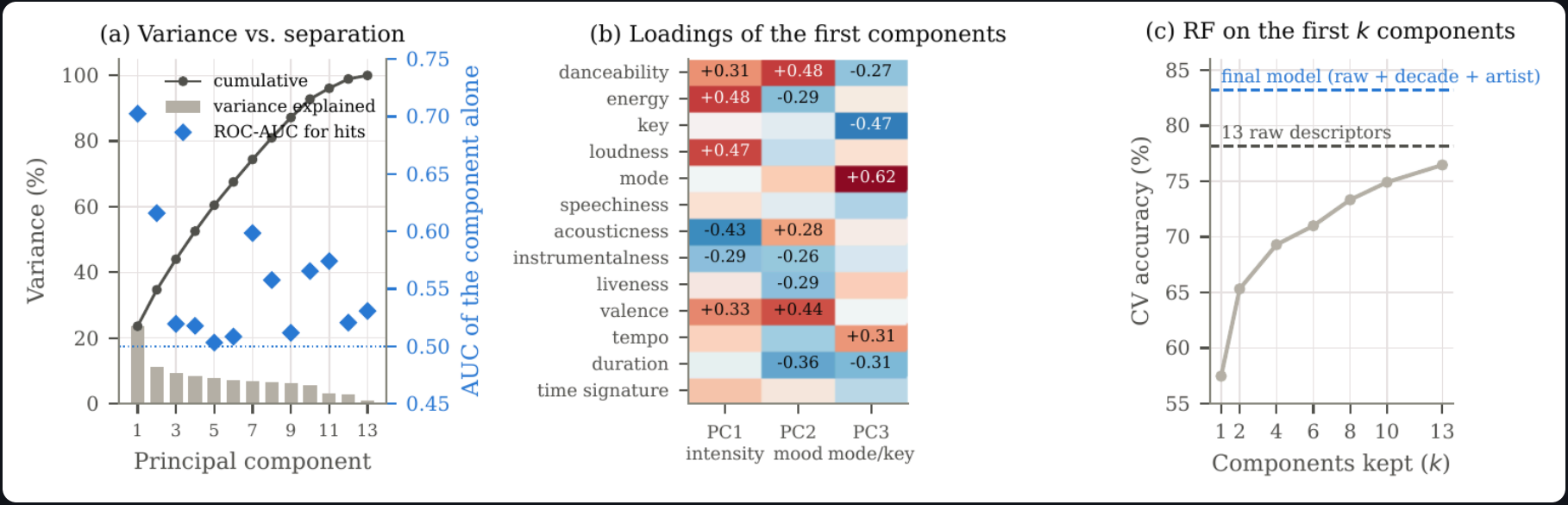


Mean over 50 random 70/30 splits · paper Table V, Fig. 2

One model per decade, 13 audio + chorus_hit + sections — exactly as in the thesis. Our re-implementation lands **on or just above** their published numbers, which validates the protocol.

Adding artist history: **+2.1 to +3.2 points** on every split; the improved models beat the published accuracy on **100 %** of splits.

Which sound numbers matter? Variance is not what separates hits



PC1 = **intensity** (energy, loudness vs acousticness), 24 % of the variance, AUC .70. PC2 = **mood** (danceability, valence), AUC .62.

PC3–PC6 hold 33 % of the variance but separate hits like a coin (AUC .50–.52).

Best single numbers: instrumentalness (.74), danceability (.70). Components instead of raw numbers: **−1.7** → keep the raw numbers.

Paper: Appendix E, Fig. 11

We tested on another dataset, built without that rule

ISMIR 2019 dataset (published, 11,662 songs): non-hits are **random songs**.

76.3%

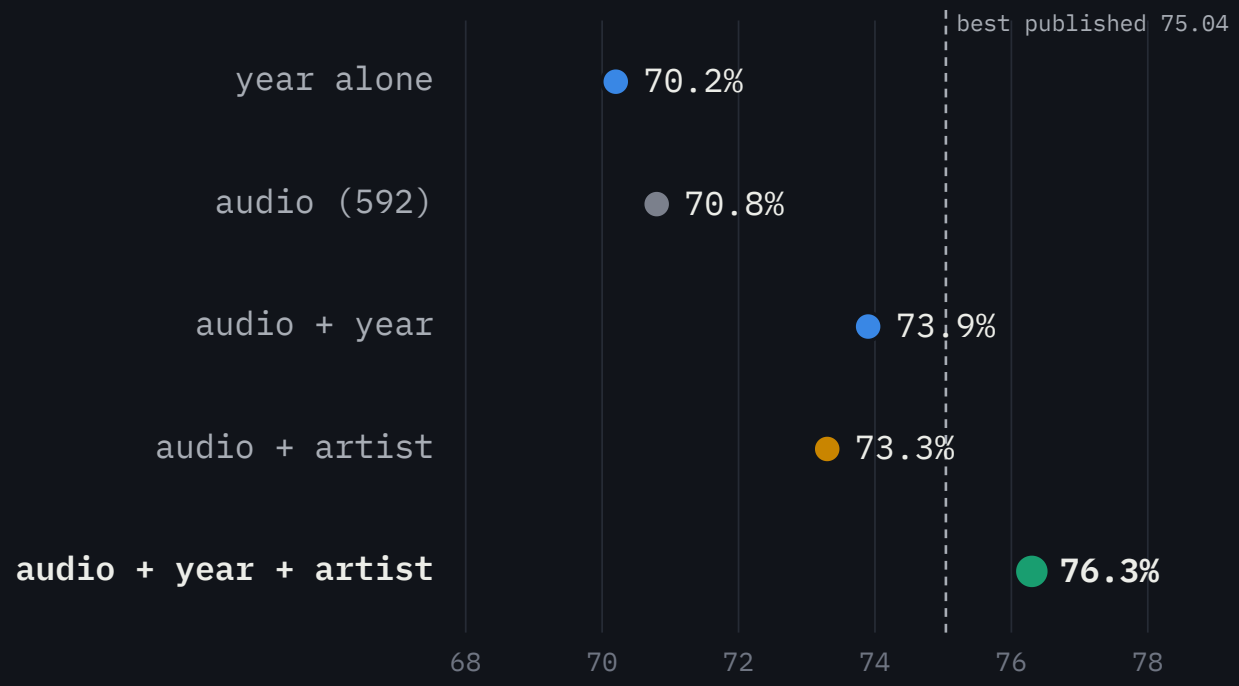
with our clue — **better than the best published result (75.0 %)**.

But another trap

The **year alone** already gets **70 %**: the hits in this dataset are older songs. The computer can win just by looking at the date.

EXTERNAL VALIDATION · ISMIR 2019, PROTOCOL A (AS IN THE ORIGINAL STUDY)

With artist history we reach 76.3 %, above the best published 75.04 %



Random Forest on 592 Essentia descriptors · 5×2 CV mean · paper Table VI

Audio + year mirrors the inputs of the best published model and gives 73.9 %. Artist history adds **+2.4 points** (10/10 folds, $p = 7{\cdot}10^{-8}$).

But look at **year alone: 70.2 %** — as good as 592 audio descriptors. Hits in this dataset are older (median 1987) than non-hits (2001).

EXTERNAL VALIDATION · SPOTGENTRACK, 101,909 TRACKS

With 81 % non-hits, accuracy hides the signal — ROC-AUC shows it



Artist history: AUC **.690 → .762** (+ decade .777). The 207 low-level audio features add nothing on top of it. Hits found: 8 % → 25 % (F1 .147 → .358). 5-fold CV · paper Table VII

Artists never seen, and brand-new artists

Artists never seen in training

We made sure no test artist appeared in training. The gain **stays the same** (+5.0): the computer does not just memorise names.

A brand-new artist (first song ever)

No history, so “who” **cannot help** (+0.4, nothing). “When” helps a little (+2.2). This is the hardest case — and the main thing left to solve.

ROBUSTNESS · FOLDS GROUPED BY ARTIST

Does the model just recognise artists? No — the gains survive when test artists are never seen

HIT PREDICTOR · decade + artist

+5.0 → +5.0

82.2 % grouped (83.2 % random). 25/25 folds, $p = 10^{-13}$.

ISMIR B · artist history

+2.6 → +2.9

10/10 folds, $p = 9{\cdot}10^{-7}$. Year still useless.

ISMIR A · audio + year + artist

75.9%

CI [75.2, 76.6] — still entirely above the published 75.04 %.

All accuracies drop by up to one point — the audio carried a little artist identity — but the gains remain (ISMIR A loses at most half a point). Paper Table VIII

ROBUSTNESS · COLD START

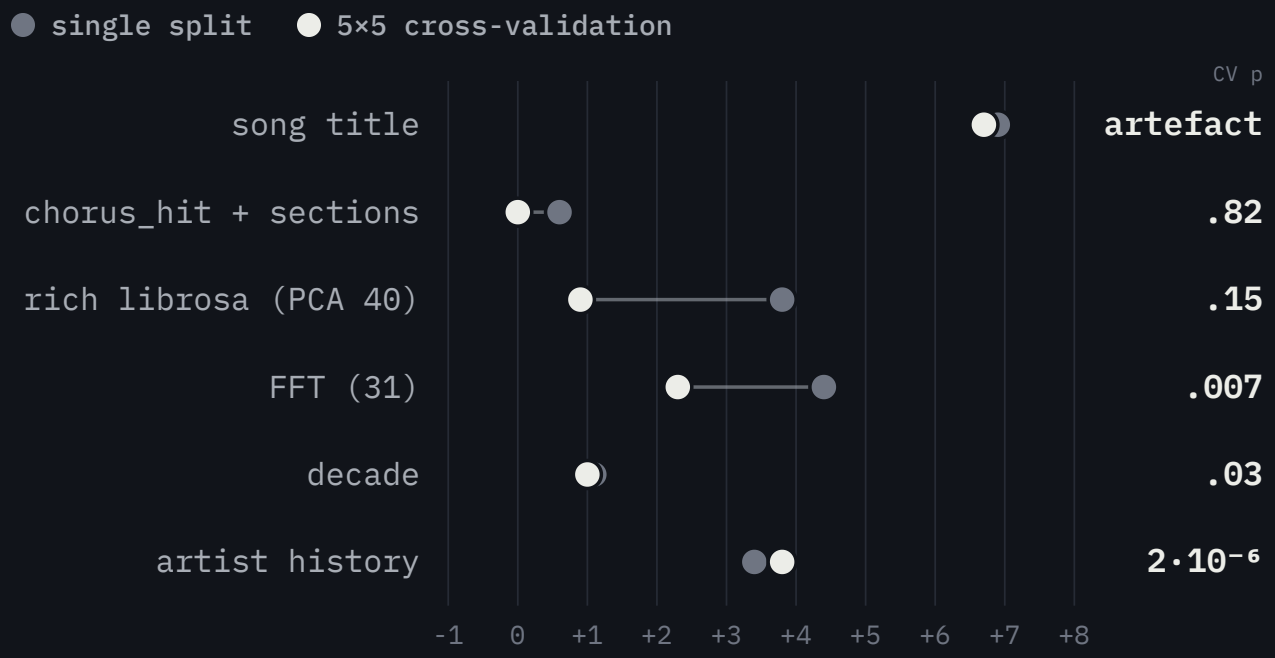
For a new artist, it is the era that helps — not the artist feature

DATASET	HITS NEW / KNOWN	NEW · + ERA	NEW · + ARTIST	KNOWN · + ARTIST
Hit Predictor	39 % / 97 %	+2.2 $p=2\cdot10^{-9}$	+0.4 n.s.	+15.0 $p=3\cdot10^{-14}$
ISMIR A	36 % / 75 %	+3.4 $p=6\cdot10^{-5}$	+0.8 n.s.	+4.4 $p=.003$
ISMIR B (year-matched)	37 % / 72 %	+0.1 n.s.	+1.4 $p=.002$	+5.3 $p=8\cdot10^{-5}$

Gain in points over audio, artist-grouped CV. On the benchmark, artist history's whole gain comes from the 17.6 % of songs by established artists.

Paper Table VIII

A big, significant gain can still be wrong: artefact or luck



Gain over audio in points · top: full benchmark · bottom: 3,562-song audio subset · paper Table IX

ARTEFACT **Song titles +6.7**, more than any musical feature. Titles don't make hits — they reveal that non-hits come from niche genres and catalogue releases.

LUCK **Rich librosa audio**: +3.8 (p = .002) on one split, **+0.9, not significant** in CV. Same for "PCA 40 is the peak" and the family knock-outs.

chorus_hit + sections, often suspected of leakage, add nothing. FFT survives here — the next slide shows why it should not.

The FFT looked like a real gain — it was recognising the genre

It looked good

The FFT splits the sound into frequencies (bass, middle, treble). **+1.4 points** on 35,000 songs, very significant.

Why it was a trap

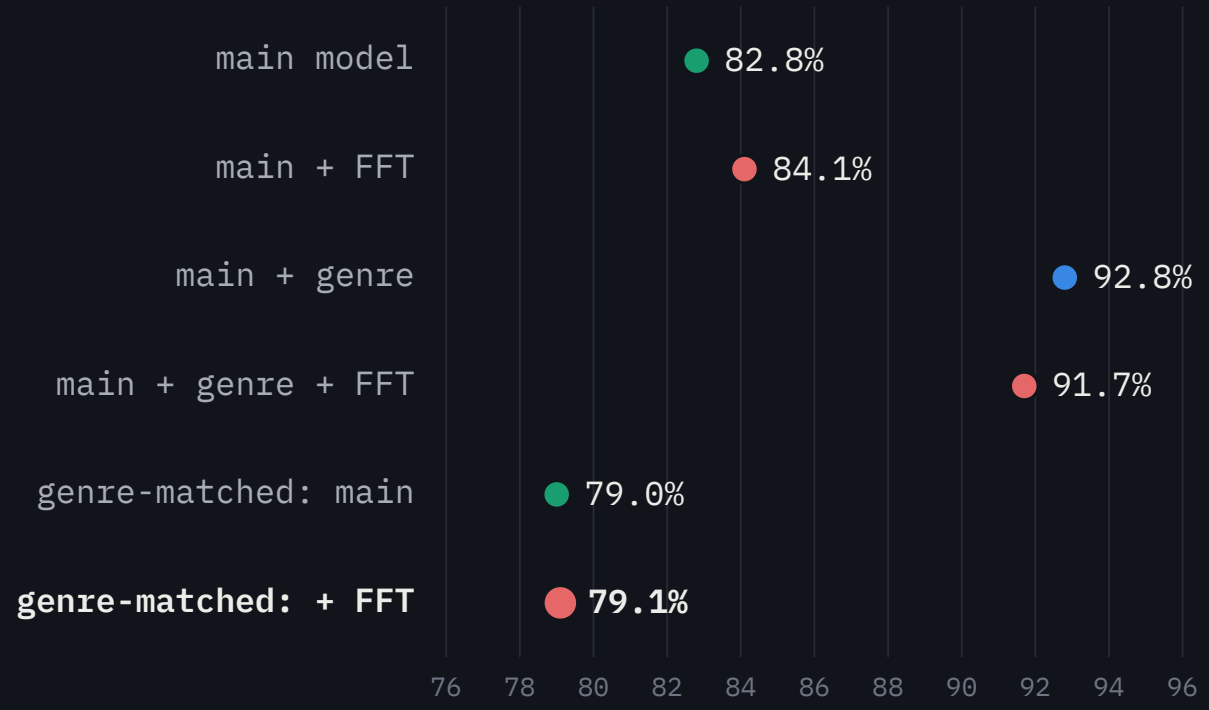
Non-hits come from niche genres (rule of the dataset), and the sound **tells the genre**. The computer was recognising jazz or Latin, not hits.

Fair test

Same number of hits and non-hits in every genre: **+0.1, nothing**.

✗ REMOVED

FFT looked like a real gain for new artists — it was the genre



20,610 songs with a clip and a Spotify genre · 5×5 CV · paper Table X

On 35,370 songs, FFT adds **+1.4** (25/25 folds, $p = 4{\cdot}10^{-9}$), all of it on new artists.

But Latin, classical, jazz, world and metal artists make **50 % of the non-hits and 1.9 % of the hits** — by construction. Audio can recognise genre (FFT: 49.9 % vs 26.1 % by chance).

With genre known, FFT **lowers** accuracy (−1.1). On a **genre-matched sample** (as many hits as non-hits in every genre): **+0.1, p = .84**. REMOVE

Genre alone lifts accuracy to 92.8 %, but Spotify assigns genres afterwards, so we use it only as a control.

Train on one decade, test k decades later: artist history ages well, FFT does not

K	PAIRS	AUDIO	+ ARTIST	+ FFT	+ ARTIST + FFT
0 (same decade)	6	.809	.837	.831	**.859**
1 decade later	5	.765	.796	.782	**.810**
2	4	.728	**.775**	.736	**.775**
3	3	.689	**.740**	.673	.728
4	2	.666	**.734**	.661	.718
5	1	.664	**.754**	.599	.706

Audio-only accuracy falls from **.809** in the same decade to **.689** three decades later: what a hit sounds like changes.

The advantage of **artist history grows** with the gap (+2.8 → +5.1 points).

FFT helps only nearby and hurts from three decades on: it encodes era-specific sound and genre.

Random Forest, 35,370 songs with a clip. Cells average 6 to 1 decade pairs, so no significance test. Paper Table XI.

Three more checks

ARTIST MATCHING

99.99%

of hits are found in the chart history. The songs that do not match are non-hits by artists who never charted — not a matching error.

Stricter matching −0.5 ($p = 4{\cdot}10^{-4}$); counting featured artists +0.1 (n.s.). KEEP the original rule.

LYRICS · SPOTGENTRACK, 79,420 TRACKS

+.023 AUC

Six lexical descriptors (word count, vocabulary richness, …) on top of decade + artist: AUC .774 → .797, accuracy .808 → .814 ($p = .002$).

Confirmed later on ISMIR with all controls (+4.4). KEEP

PRECISION · RECALL · F1 OF THE HIT CLASS

.787 → .831

F1 on the benchmark ($p = 7{\cdot}10^{-15}$): precision and recall both rise, so the gain is not a trade-off.

SpotGenTrack: audio finds 8 % of the hits; with decade + artist, **25 %** at the same precision (F1 .147 → .358).

SUMMARY · EVERY FEATURE WE TRIED

What we keep, what we removed, and why

FEATURE	RESULT	VERDICT	WHY
Artist history	+2.9 alone · +2.6 on ISMIR, years matched	KEEP	Survives every control
Era (decade / year)	+2.0 · useless once years are matched	KEEP	Helps new artists; on ISMIR only a date shortcut
Song title (TF-IDF)	+6.7, $p = 4 \cdot 10^{-20}$	REMOVE	Artefact of how non-hits were collected
Title length · chorus_hit + sections	+0.2 · 0.0, not significant	REMOVE	No signal
Rich librosa audio (185 → PCA 40)	+3.8 on one split → +0.9, n.s.	REMOVE	The single-split gain was luck
FFT spectral (31)	+1.4 → +0.1 genre-matched	REMOVE	Genre shortcut
Spotify artist genre	+10.1	CONTROL ONLY	Assigned afterwards: would leak
Low-level audio (207)	≈ 0 on top of artist history	REMOVE	Nothing beyond artist history
Lyrics (ISMIR: words + 8 statistics)	+4.4, fair test, unseen artists ($p = 3 \cdot 10^{-5}$)	KEEP	Survives every control
Market trend (vs last decade's hits)	+0.1, n.s.	REMOVE	Only carries the era (Chapter 6)

We learned to read the chart as a signal over time

3 parts of a time series

Trend: slow change over years.
Seasonality: repeats every year.
Residual: what is left.

What we found

New songs per week: mostly a **trend** (fewer until 2001, more again with streaming). Christmas songs: pure **seasonality** — the FFT finds the 52-week cycle by itself.

Forecasting next year

ARIMA and Holt-Winters are the best ✓ KEPT. Machine learning is not better here: 800 data points are too few.

Trend, seasonality, residual — and which forecasting algorithms work

COMPONENTS · 68 YEARS OF WEEKLY CHARTS

Debuts per week: mostly **trend** (strength .72): 11 in 1959, 5.5 in 2001, 11 again with streaming. Seasonality weak (.11): January is quiet. Largest residual spike: a **rule change** in December 1998.

Holiday songs on the chart: almost pure seasonality. The **FFT** finds the 52-week cycle and its harmonics (26, 17.4, 13 weeks) without being told; a Fourier fit follows the streaming-era peaks (R^2 = .94).

Notebook `timeseries/time_series_hot100.ipynb`: STL decomposition, ACF/PACF, ADF/KPSS, Fourier terms.

FORECASTING DEBUTS 12 MONTHS AHEAD · 11 ORIGINS · MASE (LOWER IS BETTER)

ARIMA(1,1,2)	1.82	KEEP
ARIMA + Fourier terms	1.82	KEEP
Holt-Winters (damped)	1.85	KEEP
SARIMA (seasonal)	1.89	REMOVE
Gradient boosting / random forest	1.94 / 1.99	REMOVE
Seasonal naive	2.32	BASELINE

The three best are indistinguishable (p = .97) and beat the seasonal naive forecast at 10 of 11 origins (p = .006). SARIMA: the series has almost no season to model. Trees: 800 points are too few, and they cannot extrapolate a trend.

TIME SERIES FOR HIT PREDICTION

After release, the first weeks predict a top-10 hit; before release, trends add nothing

AFTER RELEASE · FIRST WEEKS ON THE HOT 100 → TOP 10?

INPUT	AUC	F1
year, month, artist history only	.59	.15
first week (debut position)	.75	.40
first 2 weeks	.85	.53
first 3 weeks	**.89**	**.61**
first 4 weeks	.92	.67

31,925 songs, 16 % reach the top 10. Random Forest on positions, best rank, slope and acceleration + context; 5×3 CV, every gain 15/15 folds, $p < 10^{-6}$.

Not trivial: on songs **not yet in the top 10** after 3 weeks, AUC **.87**; trained before 2000 and tested after, **.85**. KEEP

BEFORE RELEASE · MARKET TREND ON THE KAGGLE BENCHMARK

Distance of a song's audio to the **previous decade's hits**, and whether it moves in the direction of the trend.

+0.1 pt

on top of the main model, p = .66. REMOVE

Per-feature deviation: +0.5 (p = .006, indicative). On audio alone the trend adds +1.8 — less than the decade itself (+2.0): it only carries **era**, which the model already has.

Time series help once a song is out: its first weeks are the best predictor we found. Before release, they add nothing to artist history and era.

STATISTICS · ABOUT 60 P-VALUES IN ONE STUDY

With many tests, some pass by chance — so we apply Bonferroni: $p < 0.05 / 60 \approx 8 \cdot 10^{-4}$

SURVIVES — OUR CONCLUSIONS REST ON THESE

- Main gains on the benchmark ($p = 7 \cdot 10^{-17}$)
- All five classifiers
- Head-to-head with Du, all four models
- Temporal hold-out on unseen decades
- Artist history on ISMIR, protocols A and B
- Artist-grouped gains
- The genre control ($p \leq 5 \cdot 10^{-7}$)

INDICATIVE ONLY — $8 \cdot 10^{-4} < p < 0.05$

- Decade and FFT gains on the small audio subset
- Most single-split gains on that subset
- SpotGenTrack accuracy, artist alone ($p = 10^{-3}$)
- ISMIR cold-start gains ($p = .002, .003$)
- Lyrics on SpotGenTrack ($p = .002$)

HONESTY · WHAT CHANGED SINCE OUR FIRST PRESENTATION

Eight claims did not survive re-checking

EARLIER SLIDE SAID	WHAT IS ACTUALLY TRUE
Reference paper at 0.877	Middlebrook & Sheik — a **different dataset**, not comparable
Rich audio +3.8, “PCA 40 is the peak”	**+0.9, not significant** in CV; all PCA sizes within noise
Billboard 2000–23 validation: no gain	**Invalid**: 2,911 of 3,397 rows had no audio features
SpotGenTrack “our 2 features alone” .8222	That score was **artist + low-level audio**
Tilburg .878 / .890, “a close comparison”	Exact protocol, 50 splits: **.867 / .861** vs published .84 / .82
FFT +1.4 for new artists (our own R7)	A **genre shortcut**: +0.1 on a genre-matched sample
“Artist names match for only 52 %”	**99.99 % of hits** match; the rest are non-hits by artists who never charted
Artist history “works on new songs too”	It does **not help new artists** on the benchmark (+0.4, n.s.)

WHAT IS NEW

Artist history is not a new idea. Proving what it is worth is.

ALREADY EXISTED

Interiano et al. (2018): a “superstar” indicator (did the artist chart recently?) improves a Random Forest on 500,000 UK songs.

Middlebrook & Sheik (2019) list an artist past-performance field, but never isolate or test it.

Neither tests significance, compares on the same data, or checks for sampling artefacts.

OUR CONTRIBUTION

- A **leakage-free** version (strictly earlier periods)
- A significance-tested **head-to-head** on the same data as prior work
- **External validation** with a year-matched control
- **Unseen-artist** and **cold-start** analysis
- An **audit** of artefacts and single-split luck

What this study cannot say

LIMITATIONS

- Artists matched by name, not by a canonical ID (homonyms)
- No deep-learning model: we compare inputs at equal model
- Artist history partly measures **cumulative advantage**, not a song's appeal
- ISMIR: a Random Forest and a different draw of negatives, not their network
- Du: re-drawn splits, since the original R seed cannot be replayed
- Audio clips come from Deezer, not from Spotify

NEXT

- Features that help **first-time artists**: label, collaborators, early streams — always with a genre-matched control
- Lyrics for the main dataset; meaning of lyrics; learned audio embeddings — under the same audit
- Early-trajectory forecasting (Chapter 6) with streaming data
- A benchmark whose negatives ignore artist, genre and date